

A uniform BV–flux bound for the corrected semi-discrete Brenier interface conjecture

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Status. Candidate full solution, likely valid, subject to human review. The theorem proves the corrected conjecture in Remark 6.12 of the 2025 published version of Divol–Niles-Weed–Pooladian, in its natural common-bounded-support formulation. The arXiv-v1 mass-ratio expression is separately shown to be false and was removed in the published normalization.

1 The corrected conjecture and the version issue

Let a source probability measure have density ρ on $\mathbb{R}^d$, and let

$$\mu = \sum_{j=1}^J \mu_j \delta_{y_j}, \quad \mu_j > 0,$$

be a discrete target. The semi-discrete Brenier potential is polyhedral:

$$\phi(x) = \max_{1 \leq j \leq J} \{\langle x, y_j \rangle - \psi_j\}.$$

Its gradient is y_j on the corresponding Laguerre cell L_j . For $x \in L_i$, the source defines

$$\Delta_{ij}(x) = 2(\langle x, y_i - y_j \rangle - \psi_i + \psi_j), \quad H_{ij}(t) = \{x \in L_i : \Delta_{ij}(x) = t\},$$

and

$$h_{ij}(t) = \int_{H_{ij}(t)} \rho(x) d\mathcal{H}^{d-1}(x).$$

The arXiv-v1 remark conjectured uniform boundedness of

$$Q_{v1}(\mu) = \sum_{i,j} \|y_i - y_j\| h_{ij}(0) \log(1 + \mu_j/\mu_i). \quad (1)$$

This cannot be the correct coefficient. For the uniform source on $[0, 1]$, target atoms $y_1 = 0, y_2 = 1$, and masses $(\mu_1, \mu_2) = (\delta, 1 - \delta)$, the cells meet at $x = \delta$, and $h_{12}(0) = h_{21}(0) = 1$. Hence

$$Q_{v1}(\mu) = -\log(\delta(1 - \delta)) \longrightarrow \infty \quad (\delta \downarrow 0). \quad (2)$$

The published normalization absorbs the masses into the entropic dual potential, $(\tilde{\psi}_\varepsilon)_j = (\psi_\varepsilon)_j - \varepsilon \log \mu_j$. The coefficient in the revised Remark 6.12 is instead

$$Q(\mu) = \log 2 \sum_{i,j} \|y_i - y_j\| h_{ij}(0), \quad (3)$$

and the authors conjecture that it is uniformly bounded over discrete measures.

Then it holds that, for all $\varepsilon > 0$

$$\|T_0^\mu - T_\varepsilon^\mu\|_{L^2(\rho)}^2 \leq C_0 e^{C_1 \varepsilon^\alpha} \varepsilon, \quad (6.17)$$

where the constants depend on $\rho_{\min}, \rho_{\max}, d, R, \mu_{\min}, J, \min_{i \neq j} \|y_i - y_j\|$, and on the maximum angle formed by three non aligned points among the atoms $\{y_j\}_{j=1}^J$.

Remark 6.12. Following the asymptotic results of [Altschuler et al. \(2022\)](#), we can take a limit of (6.16), resulting in the computation

$$\limsup_{\varepsilon \rightarrow \infty} \varepsilon^{-1} \|T_0^\mu - T_\varepsilon^\mu\|_{L^2(\rho)}^2 = \sum_{i,j} \frac{\|y_i - y_j\| h_{ij}(0)}{2} \int_0^\infty \left(1 + e^{u/2}\right)^{-1} du = \sum_{i,j} \|y_i - y_j\| h_{ij}(0) \log(2),$$

where we used that $t \mapsto h_{ij}(t)$ is continuous at $t = 0$ (which holds if, for instance, ρ has an upper bounded density with compact support). We conjecture that this quantity is uniformly bounded for all discrete measures.

Combined with (6.15) and Corollary 6.5, we can state and prove our main theorem for this

Figure 1: The corrected coefficient and conjecture, reproduced in Remark 6.12 of the author's thesis chapter based on the 2025 article.

2 Uniform interface theorem

Theorem 1. Suppose $0 \leq \rho \leq M$, with $\text{supp } \rho \subseteq B(0, R_x) \subset \mathbb{R}^d$, and suppose all target atoms satisfy $y_j \in B(0, R_y)$. Then

$$Q(\mu) \leq 2d\omega_d M R_y R_x^{d-1} \log 2, \quad (4)$$

where ω_d is the volume of the Euclidean unit ball. The bound is independent of J , of the masses μ_j , of the minimum atom separation, and of the combinatorics of the Laguerre diagram.

In particular, if source and target supports lie in the common ball $B(0, R)$, then

$$Q(\mu) \leq 2d\omega_d M R^d \log 2. \quad (5)$$

Proof. Repeated target atoms may be merged, so assume the y_j are distinct. The function ϕ is convex and globally R_y -Lipschitz. Its gradient $T = \nabla \phi$ is a piecewise-constant BV map whose values on the interiors of the Laguerre cells are the y_j .

Let F_{ij} be the relative interior of a common codimension-one facet of L_i and L_j . Direct the unit normal from L_i to L_j . Since L_i is the side on which $\langle x, y_i - y_j \rangle - \psi_i + \psi_j \geq 0$, this normal is

$$\nu_{ij} = \frac{y_j - y_i}{\|y_j - y_i\|}.$$

The BV jump formula therefore gives

$$D(\nabla \phi)|_{F_{ij}} = (y_j - y_i) \otimes \nu_{ij} \mathcal{H}^{d-1}|_{F_{ij}}.$$

Taking the trace shows that the distributional Laplacian of ϕ has facet density

$$\Delta\phi|_{F_{ij}} = \|y_i - y_j\| \mathcal{H}^{d-1}|_{F_{ij}}. \quad (6)$$

Lower-dimensional multiple-tie sets carry no $\mathcal{H}^{d-1}$ mass. Inactive affine pieces can be discarded. Thus (6) accounts for the whole Laplacian measure of the polyhedral convex function.

At $t = 0$, $H_{ij}(0)$ is precisely the trace of the common facet on L_i . The ordered sum counts every facet twice. Therefore, for any $r > R_x$,

$$\begin{aligned} \sum_{i,j} \|y_i - y_j\| h_{ij}(0) &= 2 \sum_{i < j} \|y_i - y_j\| \int_{F_{ij}} \rho d\mathcal{H}^{d-1} \\ &\leq 2M \Delta\phi(B(0, r)). \end{aligned} \quad (7)$$

It remains only to bound the total Laplacian mass. Let $\phi_\eta = \phi * \zeta_\eta$ be a standard mollification. Convexity and the gradient bound are preserved: $\|\nabla\phi_\eta\| \leq R_y$. By the divergence theorem,

$$\begin{aligned} \Delta\phi_\eta(B(0, r)) &= \int_{\partial B(0, r)} \nabla\phi_\eta \cdot n d\mathcal{H}^{d-1} \\ &\leq R_y \mathcal{H}^{d-1}(\partial B(0, r)) = d\omega_d R_y r^{d-1}. \end{aligned} \quad (8)$$

The nonnegative measures $\Delta\phi_\eta$ converge weakly to $\Delta\phi$. For all but countably many radii the boundary has zero $\Delta\phi$ -mass, so (8) passes to the limit. Choosing such radii $r \downarrow R_x$ and using (7) yields

$$\sum_{i,j} \|y_i - y_j\| h_{ij}(0) \leq 2d\omega_d M R_y R_x^{d-1}.$$

Multiplication by $\log 2$ proves (4), and (5) follows by taking $R_x = R_y = R$. $\square$

3 Interpretation and checks

The proof identifies the conjectured interface sum with twice a weighted part of the trace measure of $D^2\phi$. Convexity makes this Hessian measure positive, while the bound on the slopes of ϕ converts its total mass into boundary flux. This is why neither small target masses nor complicated Laguerre diagrams cause divergence in the corrected expression.

In dimension one, the identity reduces to the elementary statement that the sum of successive jumps of a monotone step function is its total variation. For the two-atom example used in (2), the corrected quantity is always $2\log 2$, independently of δ .

The common target-support bound is necessary. If “all discrete measures” is interpreted literally with no common bound on the atom locations, scaling two target atoms scales Q linearly. Theorem 1 resolves the fixed-ball formulation used in the stability results of the source paper.

4 Novelty audit and reviewer focus

The run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:2404.02855, the exact paper title, the corrected expression, Laguerre-interface sums, BV variation, and Brenier-potential flux; no duplicate was found. Bounded literature searches on 9 August 2026 used the exact conjecture sentence and formula and checked later work on semi-discrete entropic maps. The corrected conjecture remains present in the published article’s thesis reproduction, and no later resolution was located.

The main reviewer checks are the factor of two in (7), the treatment of degenerate multiple ties, and the intended common-support interpretation of “uniformly bounded.” The first follows because the source sums ordered pairs; the second is the standard BV jump decomposition after inactive pieces are removed. Novelty confidence is moderate because the underlying BV and divergence facts are classical, though their application here seems unrecorded.

References

- [1] V. Divol, J. Niles-Weed, and A.-A. Pooladian, *Tight Stability Bounds for Entropic Brenier Maps*, International Mathematics Research Notices, 2025(7), rnaf078 (2025), doi:10.1093/imrn/rnaf078; arXiv:2404.02855.
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- [3] L. Ambrosio, N. Fusco, and D. Pallara, *Functions of Bounded Variation and Free Discontinuity Problems*, Oxford University Press, 2000.